Spectral stability of non-transonic plasma flows in the magnetic nozzle, revisited with the infinite-dimensional Beyn method

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Abstract

The polynomial eigenvalue problem of the thesis [1] for plasma flow in a magnetic nozzle is solved again for the four non-transonic configurations (constant and spatially varying velocity, subsonic and supersonic, each with Dirichlet and fixed-open boundary conditions) using InfBeyn, the contour-integral method of Colbrook and Townsend [2] applied directly to the differential operator. The method returns every eigenvalue in a prescribed window of the complex plane, and each one is certified through the pseudospectral quantity γ_n. There are no spurious eigenvalues and no convergence test is required. The stability verdicts of the thesis are confirmed: both subsonic and supersonic flows are stable with Dirichlet boundaries, the supersonic flow is unstable with fixed-open boundaries, and the subsonic flow with a varying velocity profile has a single unstable ground mode with fixed-open boundaries. The new eigenvalues are compared with the thesis-era finite-difference, spectral-collocation and spectral-Galerkin results. For the constant-velocity problems InfBeyn agrees with the exact eigenvalues to 10^{-13}. Finite differences on $N = 101$ and $N = 201$ points miss the eigenvalues by 10^{-3} to 10^{0} and produce up to eight spurious eigenvalues in the window; collocation agrees with InfBeyn to 10^{-13} for constant velocity, to 10^{-11} for the varying profiles with Dirichlet boundaries and to 10^{-3} with fixed-open boundaries.

1 Problem

The linearized flow equations of the thesis reduce to the quadratic eigenvalue problem (thesis Eq. 3.6)

$$T(\omega)\tilde{v} \equiv \omega^2\tilde{v}+2i\omega\left(v_0\tilde{v}'+v_0'\tilde{v}\right)+\left[(1-v_0^2)\tilde{v}''-\left(3v_0+\frac{1}{v_0}\right)v_0'\tilde{v}'-\left(1-\frac{1}{v_0^2}\right)v_0'^2\tilde{v}-\left(v_0+\frac{1}{v_0}\right)v_0''\tilde{v}\right]=0 \tag{1}$$

on $z \in [-1,1]$, with either Dirichlet boundaries $\tilde{v}(-1) = \tilde{v}(1) = 0$ or fixed-open boundaries $\tilde{v}(-1) = \tilde{v}'(1) = 0$. A mode grows in time when $\operatorname{Im}\omega > 0$. The equilibrium velocity is (thesis Eq. 2.33)

$$v_0(z) = \left[-W_k\left(-\frac{B(z)^2}{B_m^2}v_m^2 e^{-v_m^2}\right)\right]^{1/2}, \qquad B(z) = B_0\left(1+R\,e^{-(z/\Delta)^2}\right), \tag{2}$$

with $B_0 = 1$, $R = 1.5$, $\Delta = 1/3$, $B_m = B(0)$, and the Lambert branch $k = 0$ for the subsonic profile ($v_m = 0.5$) and $k = -1$ for the supersonic profile ($v_m = 1.5$). The derivatives v_0', v_0'' are evaluated exactly from the equilibrium relation $(v_0^2-1)v_0' = -v_0B'/B$. The constant-velocity cases use $v_0 \equiv v_m$. Figure 1 shows the four profiles.

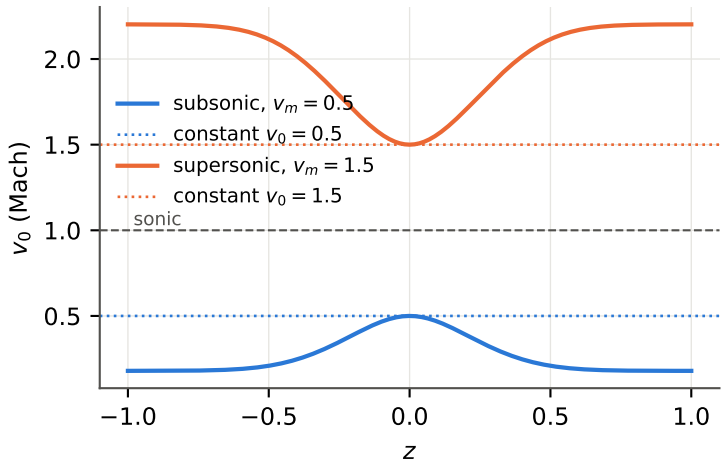


Figure 1: Equilibrium velocity profiles of the eight cases: constant ($v_0 = 0.5,\ 1.5$) and spatially varying ($v_m = 0.5,\ 1.5$ at the throat $z = 0$).

For constant velocity the eigenvalues are known in closed form (thesis Eqs. 3.10 and 3.12):

$$\text{Dirichlet:}\quad \omega_n = \frac{n\pi}{2}(1 - v_0^2), \qquad n = \pm 1, \pm 2, \ldots \tag{3}$$

$$\text{fixed-open:}\quad \omega_n = (v_0^2 - 1)\left[\phi_n - \frac{i}{4}\ln\left|\frac{v_0 - 1}{v_0 + 1}\right|\right], \qquad \phi_n = \begin{cases} n\pi/2, & v_0 > 1, \\ (2n+1)\pi/4, & v_0 < 1. \end{cases} \tag{4}$$

Two remarks on Eq. (4). The condition $\tilde{v}'(1) = 0$ gives $e^{-4i\omega/(v_0^2-1)} = (1 + v_0)/(v_0 - 1)$; for $v_0 < 1$ the right-hand side is negative, so its logarithm carries an extra $i\pi$ and the real parts sit at half-integer multiples of $\pi(1-v_0^2)/2$. Thesis Eq. (3.12) writes $n\pi/2$ for both signs, but the values plotted in thesis Fig. 3.2(a) ($\mathrm{Re}\,\omega = -0.589, -1.767, \ldots$) already correspond to the half-integer phases ϕ_n used here. Secondly, $n = 0$ in Eq. (3) gives $\omega = 0$ with $\tilde{v} \equiv 0$ (thesis Eq. 3.9), so it is not an eigenvalue and is excluded.

2 Method

InfBeyn [2] is Beyn's contour method applied to the operator $T(\omega)$ itself. For a closed contour Γ one forms the quasimatrices

$$A_0 = \frac{1}{2\pi i}\oint_\Gamma T(z)^{-1} V\, dz, \qquad A_1 = \frac{1}{2\pi i}\oint_\Gamma z\, T(z)^{-1} V\, dz, \tag{5}$$

where V is a set of random smooth test functions. Each $T(z)^{-1}V$ is a linear boundary value problem, solved by Chebyshev collocation with the number of points doubled until the Chebyshev coefficients of the solution have decayed below 10^{-12}. The eigenvalues inside Γ are those of the small matrix pencil obtained from the SVD of A_0. Only linear BVPs are ever discretized, each to its own tolerance, so there is no global discretization of the eigenvalue problem and hence no spectral pollution.

The implementation (`inf_beyn.py`) was used as follows for every case:

- `scan()` over the window $\mathrm{Re}\,\omega \in [-0.5, 26]$, $\mathrm{Im}\,\omega \in [-3, 3]$ by adaptive rectangular subdivision (Clenshaw–Curtis rule with 33 nodes per edge, up to 7 levels, 24 test functions). A tile is

accepted only if the singular values of A_0 show a gap of at least 10^4 and the nested half rule reproduces the eigenvalues. Accepted estimates are polished by Newton's method on the bordered system $[T(\omega)u;\, c^*u - 1] = 0$ on a grid doubled until convergence.

- `verify()`: for each eigenvalue, $\gamma_n(\omega) = \min_u \|T(\omega)u\|/\|u\|$ over polynomials of degree $n = 64, 128, 256$ satisfying the boundary conditions. By Theorem 2.2 of [2] ω lies in the $\gamma_n(\omega)$-pseudospectrum of T; a true, well-conditioned eigenvalue has $\gamma_n(\omega)$ at rounding level and a much larger "halo" value of γ_n on a circle (radius 0.25) around it.
- `pseudospectra()`: γ_{128} on a 265×61 grid of the window.

The old technique of the thesis is also rerun on the same operator: second-order central finite differences on $N = 101$ and $N = 201$ equispaced points, with $\tilde{v}'(1) = 0$ imposed by a one-sided second-order stencil, linearized to a generalized eigenvalue problem (`nozzle_infbeyn.py`, function `finite_difference`). The spectral-collocation ($N = 101$ Chebyshev points) and spectral-Galerkin (30 compact Legendre combinations) eigenvalues are taken unchanged from the thesis data files in `../data/`. These three contain all eigenvalues of the discretized problem, spurious ones included; the thesis removed the spurious ones by a convergence test. Here an old eigenvalue in the window farther than 0.1 from every InfBeyn eigenvalue is counted as spurious.

Running all eight cases takes about two and a half minutes on 12 cores (forked workers); the scans themselves take 2 to 9 seconds each (Table 1).

3 Results

case	BC	#eig	BVP solves	scan [s]	max $\mathrm{Im}\,\omega$ (mode)	verdict
constant $v_0 = 0.5$	Dirichlet	22	5062	3	$0.0000\ (8.247 + 0.000i)$	stable
constant $v_0 = 0.5$	fixed-open	22	1858	2	$-0.2060\ (18.261 - 0.206i)$	stable
constant $v_0 = 1.5$	Dirichlet	13	1167	2	$0.0000\ (25.525 + 0.000i)$	stable
constant $v_0 = 1.5$	fixed-open	14	790	2	$0.5029\ (-0.000 + 0.503i)$	unstable
$v_m = 0.5$	Dirichlet	18	3617	8	$0.0000\ (5.718 + 0.000i)$	stable
$v_m = 0.5$	fixed-open	18	3680	9	$0.0388\ (0.647 + 0.039i)$	unstable
$v_m = 1.5$	Dirichlet	6	413	3	$0.0000\ (24.683 + 0.000i)$	stable
$v_m = 1.5$	fixed-open	7	413	8	$1.0136\ (0.000 + 1.014i)$	unstable

Table 1: Summary of the InfBeyn scans. Eigenvalue counts are for the window $\mathrm{Re}\,\omega \in [-0.5, 26]$, $\mathrm{Im}\,\omega \in [-3, 3]$; the growth rate is the largest $\mathrm{Im}\,\omega$ found and the mode that attains it.

Every eigenvalue of every case passed the certification: $\gamma_{256}(\omega) \le 3 \times 10^{-8}$ (at the rounding floor of the computation) with a halo larger by seven to ten orders of magnitude, and the pseudospectral plots show no region of small γ_n other than the isolated discs around the eigenvalues. No tile was left unresolved. The stability verdicts of the thesis are reproduced in all eight cases.

3.1 Constant velocity

Subsonic, $v_0 = 0.5$. With Dirichlet boundaries the 22 eigenvalues in the window are real and equal to $n\pi(1 - v_0^2)/2$ to within 10^{-13} (Fig. 2, Table 2). With fixed-open boundaries all 22 eigenvalues share the damping rate $\mathrm{Im}\,\omega = -0.2060 = \frac{3}{16}\ln\frac{1}{3}$ (Fig. 3, Table 3); the flow is stable.

Supersonic, $v_0 = 1.5$. Dirichlet: 13 real eigenvalues $n\pi(v_0^2-1)/2$, stable (Fig. 4, Table 4). Fixed-open: 14 eigenvalues with the common growth rate $\mathrm{Im}\,\omega = 0.50295 = \frac{5}{16}\ln 5$, including the purely imaginary $n = 0$ mode; the flow is unstable (Fig. 5, Table 5). The $n = 0$ eigenvalue is matched to 2×10^{-11}, the others to 10^{-12}; its eigenfunction is the smoothest but the resolvent is largest there.

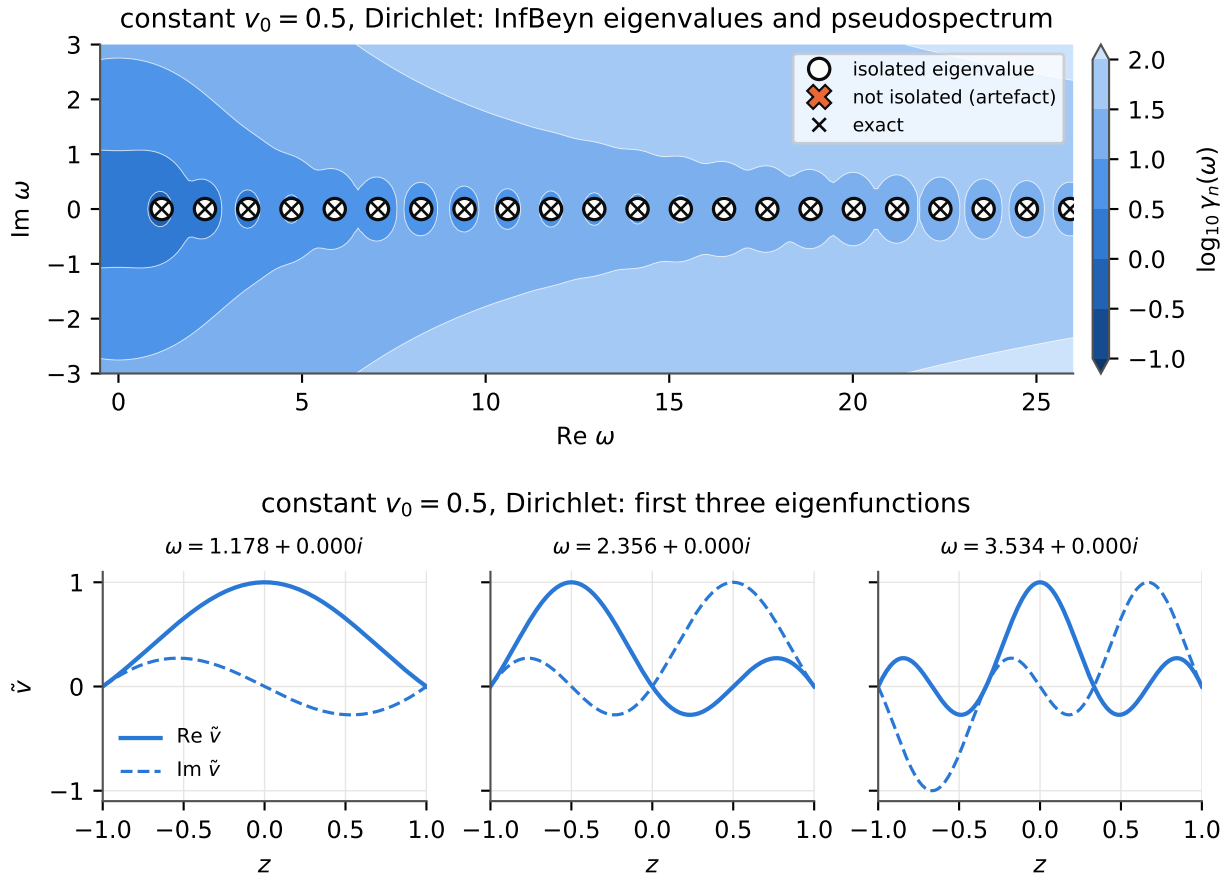


Figure 2: Constant $v_0 = 0.5$, Dirichlet. Top: InfBeyn eigenvalues (circles) on the pseudospectrum $\log_{10}\gamma_{128}(\omega)$; crosses are the exact eigenvalues. Bottom: first three eigenfunctions (Chebyshev series returned by the method, normalized so the largest value is 1).

n	ω_n (InfBeyn)	$\gamma_{256}(\omega_n)$	halo	polished	$\lvert\omega_n - \omega_n^{\mathrm{exact}}\rvert$
1	$1.17809725 + 0.00000000i$	6.5×10^{-10}	7.0×10^{-1}	yes	2.9×10^{-14}
2	$2.35619449 + 0.00000000i$	6.5×10^{-10}	1.5×10^{0}	yes	3.5×10^{-14}
3	$3.53429174 + 0.00000000i$	6.2×10^{-10}	2.3×10^{0}	yes	7.8×10^{-14}
4	$4.71238898 - 0.00000000i$	7.0×10^{-10}	3.1×10^{0}	yes	2.4×10^{-14}
5	$5.89048623 - 0.00000000i$	6.4×10^{-10}	3.8×10^{0}	yes	3.6×10^{-14}
6	$7.06858347 - 0.00000000i$	6.8×10^{-10}	4.6×10^{0}	yes	4.1×10^{-14}
7	$8.24668072 + 0.00000000i$	6.9×10^{-10}	5.4×10^{0}	yes	2.0×10^{-14}
8	$9.42477796 - 0.00000000i$	6.5×10^{-10}	6.2×10^{0}	yes	2.0×10^{-14}
9	$10.60287521 - 0.00000000i$	6.1×10^{-10}	6.9×10^{0}	yes	1.4×10^{-14}
10	$11.78097245 - 0.00000000i$	6.9×10^{-10}	7.7×10^{0}	yes	3.1×10^{-14}

Table 2: Constant $v_0 = 0.5$, Dirichlet: first ten eigenvalues, certification and error.

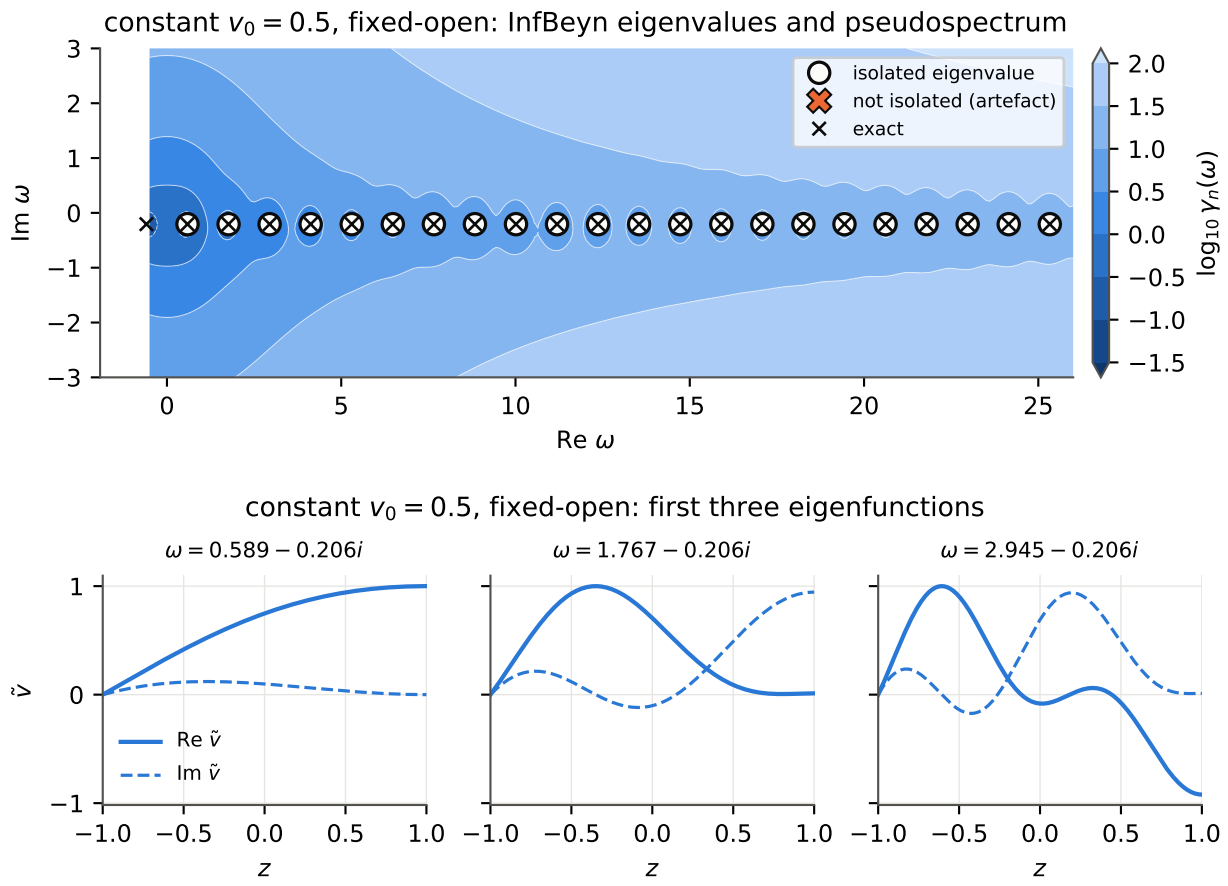


Figure 3: Constant $v_0 = 0.5$, fixed-open.

n	ω_n (InfBeyn)	$\gamma_{256}(\omega_n)$	halo	polished	$\lvert\omega_n - \omega_n^{\text{exact}}\rvert$
1	$0.58904862 - 0.20598980i$	9.0×10^{-10}	2.7×10^{-1}	yes	4.1×10^{-12}
2	$1.76714587 - 0.20598980i$	7.0×10^{-10}	8.3×10^{-1}	yes	1.8×10^{-12}
3	$2.94524311 - 0.20598980i$	9.1×10^{-10}	1.4×10^{0}	yes	9.6×10^{-13}
4	$4.12334036 - 0.20598980i$	8.4×10^{-10}	2.0×10^{0}	yes	2.4×10^{-14}
5	$5.30143760 - 0.20598980i$	7.0×10^{-10}	2.6×10^{0}	yes	9.0×10^{-13}
6	$6.47953485 - 0.20598980i$	7.4×10^{-10}	3.2×10^{0}	yes	3.0×10^{-14}
7	$7.65763209 - 0.20598980i$	7.5×10^{-10}	3.8×10^{0}	yes	5.9×10^{-13}
8	$8.83572934 - 0.20598980i$	7.6×10^{-10}	4.4×10^{0}	yes	1.7×10^{-13}
9	$10.01382658 - 0.20598980i$	8.5×10^{-10}	5.0×10^{0}	yes	3.2×10^{-13}
10	$11.19192383 - 0.20598980i$	7.9×10^{-10}	5.6×10^{0}	yes	4.0×10^{-13}

Table 3: Constant $v_0 = 0.5$, fixed-open: first ten eigenvalues. The exact values are Eq. (4) with the half-integer phases.

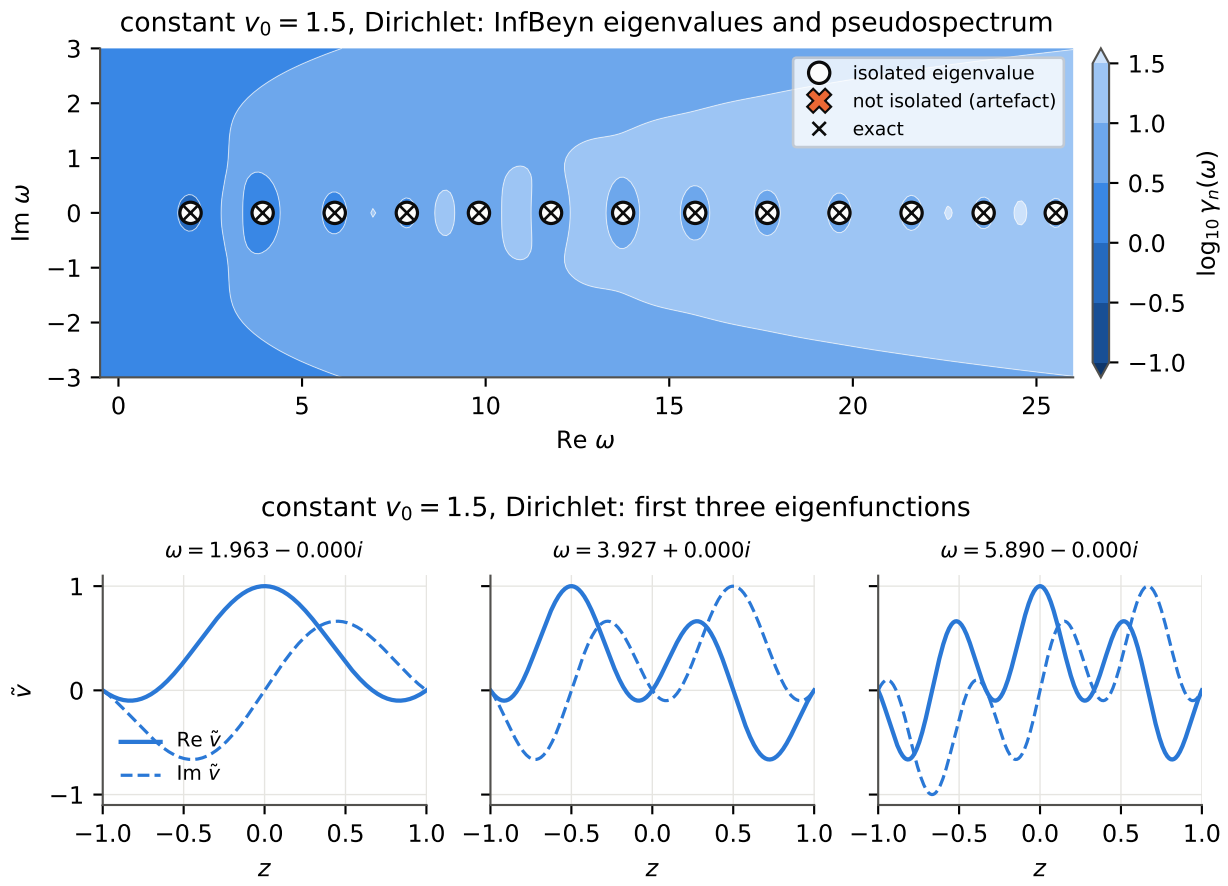


Figure 4: Constant $v_0 = 1.5$, Dirichlet.

n	ω_n (InfBeyn)	$\gamma_{256}(\omega_n)$	halo	polished	$\lvert\omega_n - \omega_n^{\text{exact}}\rvert$
1	$1.96349541 - 0.00000000i$	2.1×10^{-9}	7.4×10^{-1}	yes	3.0×10^{-13}
2	$3.92699082 + 0.00000000i$	1.9×10^{-9}	1.5×10^{0}	yes	5.8×10^{-14}
3	$5.89048623 - 0.00000000i$	2.2×10^{-9}	2.2×10^{0}	yes	1.1×10^{-13}
4	$7.85398163 - 0.00000000i$	2.1×10^{-9}	3.0×10^{0}	yes	2.8×10^{-14}
5	$9.81747704 + 0.00000000i$	2.0×10^{-9}	3.7×10^{0}	yes	7.1×10^{-14}
6	$11.78097245 - 0.00000000i$	1.8×10^{-9}	4.4×10^{0}	yes	3.5×10^{-14}
7	$13.74446786 - 0.00000000i$	2.0×10^{-9}	5.2×10^{0}	yes	7.6×10^{-14}
8	$15.70796327 - 0.00000000i$	1.8×10^{-9}	5.9×10^{0}	yes	5.2×10^{-14}
9	$17.67145868 - 0.00000000i$	2.2×10^{-9}	6.7×10^{0}	yes	3.0×10^{-13}
10	$19.63495408 + 0.00000000i$	2.1×10^{-9}	7.4×10^{0}	yes	2.3×10^{-13}

Table 4: Constant $v_0 = 1.5$, Dirichlet: first ten eigenvalues.

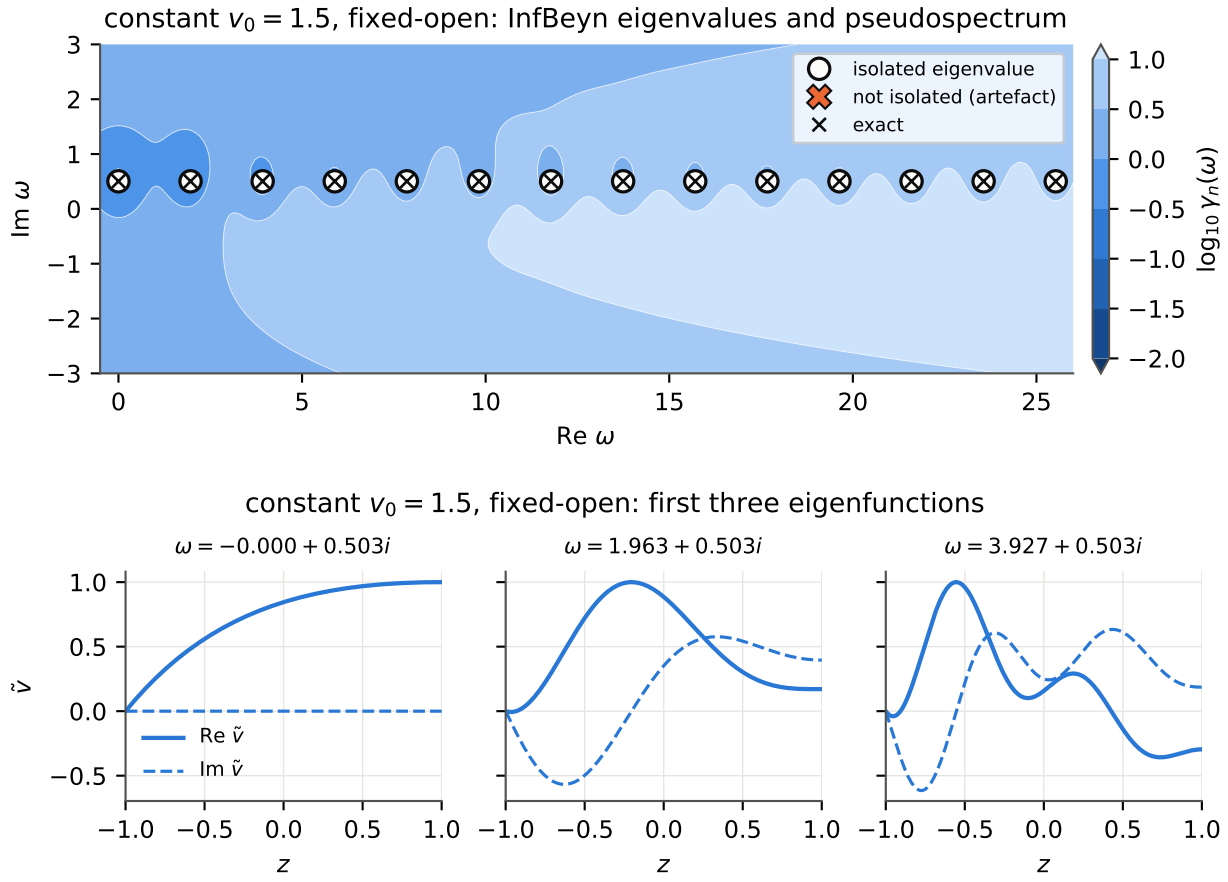


Figure 5: Constant $v_0 = 1.5$, fixed-open. All modes grow at the same rate $\mathrm{Im}\,\omega = 0.503$.

n	ω_n (InfBeyn)	$\gamma_{256}(\omega_n)$	halo	polished	$\lvert\omega_n - \omega_n^{\mathrm{exact}}\rvert$
1	$-0.00000000 + 0.50294935i$	1.0×10^{-9}	3.4×10^{-1}	yes	2.1×10^{-11}
2	$1.96349541 + 0.50294935i$	7.5×10^{-10}	4.0×10^{-1}	yes	1.7×10^{-12}
3	$3.92699082 + 0.50294935i$	6.1×10^{-10}	6.8×10^{-1}	yes	1.2×10^{-12}
4	$5.89048623 + 0.50294935i$	7.3×10^{-10}	9.9×10^{-1}	yes	9.3×10^{-13}
5	$7.85398163 + 0.50294935i$	9.2×10^{-10}	1.3×10^{0}	yes	9.5×10^{-14}
6	$9.81747704 + 0.50294935i$	8.9×10^{-10}	1.6×10^{0}	yes	3.0×10^{-13}
7	$11.78097245 + 0.50294935i$	8.1×10^{-10}	1.9×10^{0}	yes	1.6×10^{-13}
8	$13.74446786 + 0.50294935i$	7.9×10^{-10}	2.2×10^{0}	yes	1.0×10^{-12}
9	$15.70796327 + 0.50294935i$	6.8×10^{-10}	2.6×10^{0}	yes	8.7×10^{-13}
10	$17.67145868 + 0.50294935i$	7.9×10^{-10}	2.9×10^{0}	yes	8.3×10^{-13}

Table 5: Constant $v_0 = 1.5$, fixed-open: first ten eigenvalues.

3.2 Spatially varying velocity

Subsonic, $v_m = 0.5$. Dirichlet: 18 real eigenvalues, the first being $\omega_1 = 1.0562$; the flow is stable (Fig. 6, Table 6). Fixed-open: the ground mode $\omega_1 = 0.64662 + 0.03884i$ is unstable, the second mode is strongly damped ($\mathrm{Im}\,\omega_2 = -0.2709$), and all higher modes are damped with $\mathrm{Im}\,\omega \to -0.0825$ (Fig. 7, Table 7). This is exactly the "stable except the ground mode" picture of thesis Fig. 4.17; the growth rate of the ground mode is now certified rather than inferred from the

agreement of two discretizations.

Supersonic, $v_m = 1.5$**.** The spectrum is much sparser: the eigenvalues are spaced by about 4.1 instead of 2.0 for the constant profile, because the mean of $v_0^2 - 1$ over the nozzle is larger ($v_0 \approx 2.2$ at the ends). Dirichlet: 6 real eigenvalues $3.4746, 8.3967, 12.5102, \ldots$, stable (Fig. 8, Table 8). Fixed-open: 7 eigenvalues, all with $\mathrm{Im}\,\omega$ between 0.52 and 1.01; the purely imaginary mode $\omega_1 = 1.01357i$ has the largest growth rate (Fig. 9, Table 9). The flow is unstable.

The eigenvalue $\omega_1 = 1.01357i$ is the only one flagged "not polished": its eigenfunction (left panel of Fig. 9) needs more than 2000 Chebyshev coefficients to reach the 10^{-12} tail, so Newton's method stalls at the rounding floor with a residual real part of 10^{-8}. An independent InfBeyn solve on a circle of radius 0.2 around it returns $\omega = 1.0135697i$ with real part 2×10^{-14} and a clean rank-one singular-value gap, and $\gamma_{256}(\omega) = 3 \times 10^{-8}$ with halo 0.38; the eigenvalue is genuine and accurate to about 10^{-7}. Its real part vanishes for the same reason as in the constant case: the $\mathrm{Re}\,\omega = 0$ mode is the non-oscillating one.

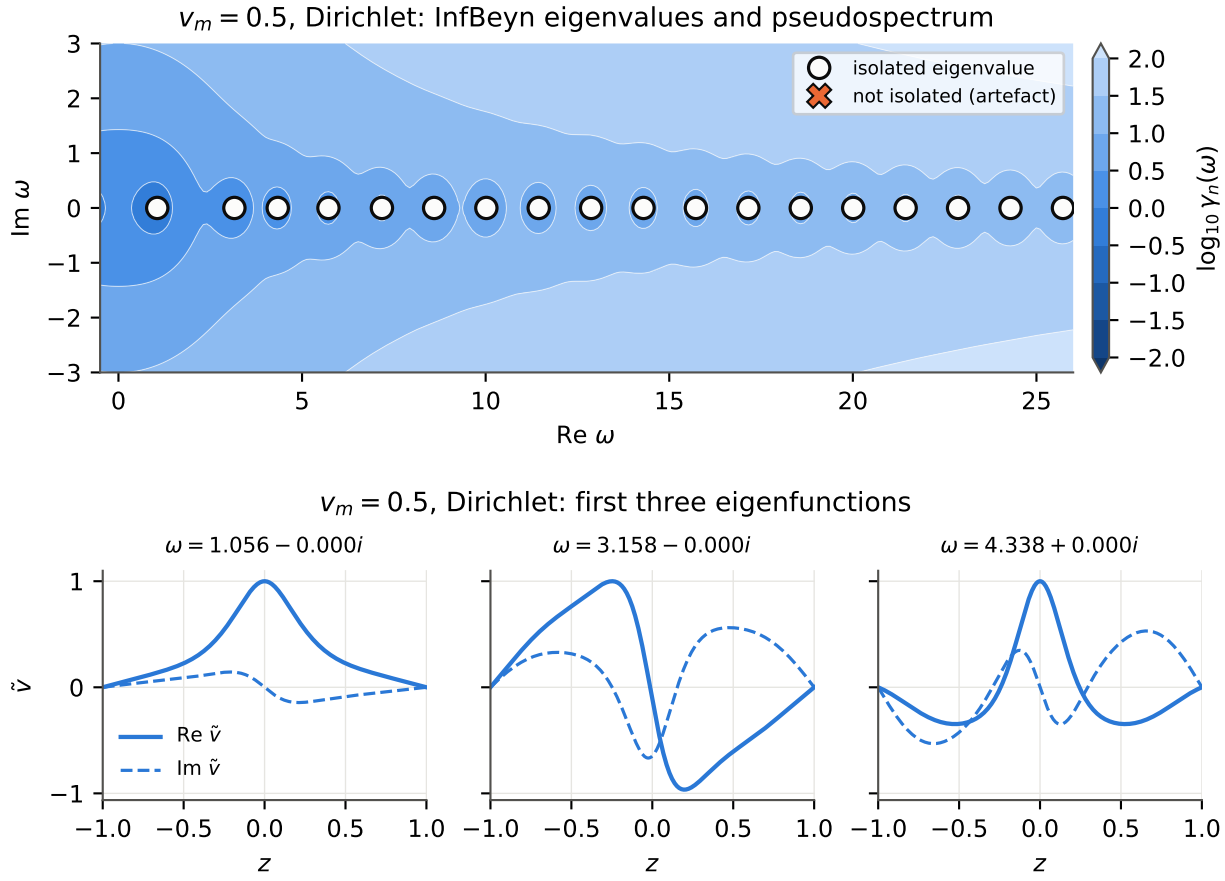


Figure 6: Varying profile, $v_m = 0.5$, Dirichlet.

n	ω_n (InfBeyn)	$\gamma_{256}(\omega_n)$	halo	polished
1	$1.05621525 - 0.00000000i$	1.4×10^{-9}	4.7×10^{-1}	yes
2	$3.15772808 - 0.00000000i$	1.3×10^{-9}	1.4×10^{0}	yes
3	$4.33790802 + 0.00000000i$	1.2×10^{-9}	2.0×10^{0}	yes
4	$5.71847609 + 0.00000000i$	1.3×10^{-9}	2.6×10^{0}	yes
5	$7.17529391 - 0.00000000i$	1.5×10^{-9}	3.3×10^{0}	yes
6	$8.59747157 + 0.00000000i$	1.4×10^{-9}	3.9×10^{0}	yes
7	$10.01903504 - 0.00000000i$	1.5×10^{-9}	4.6×10^{0}	yes
8	$11.44531362 + 0.00000000i$	1.5×10^{-9}	5.3×10^{0}	yes
9	$12.87226584 - 0.00000000i$	1.5×10^{-9}	5.9×10^{0}	yes
10	$14.29929220 + 0.00000000i$	1.4×10^{-9}	6.6×10^{0}	yes

Table 6: $v_m = 0.5$, Dirichlet: first ten eigenvalues.

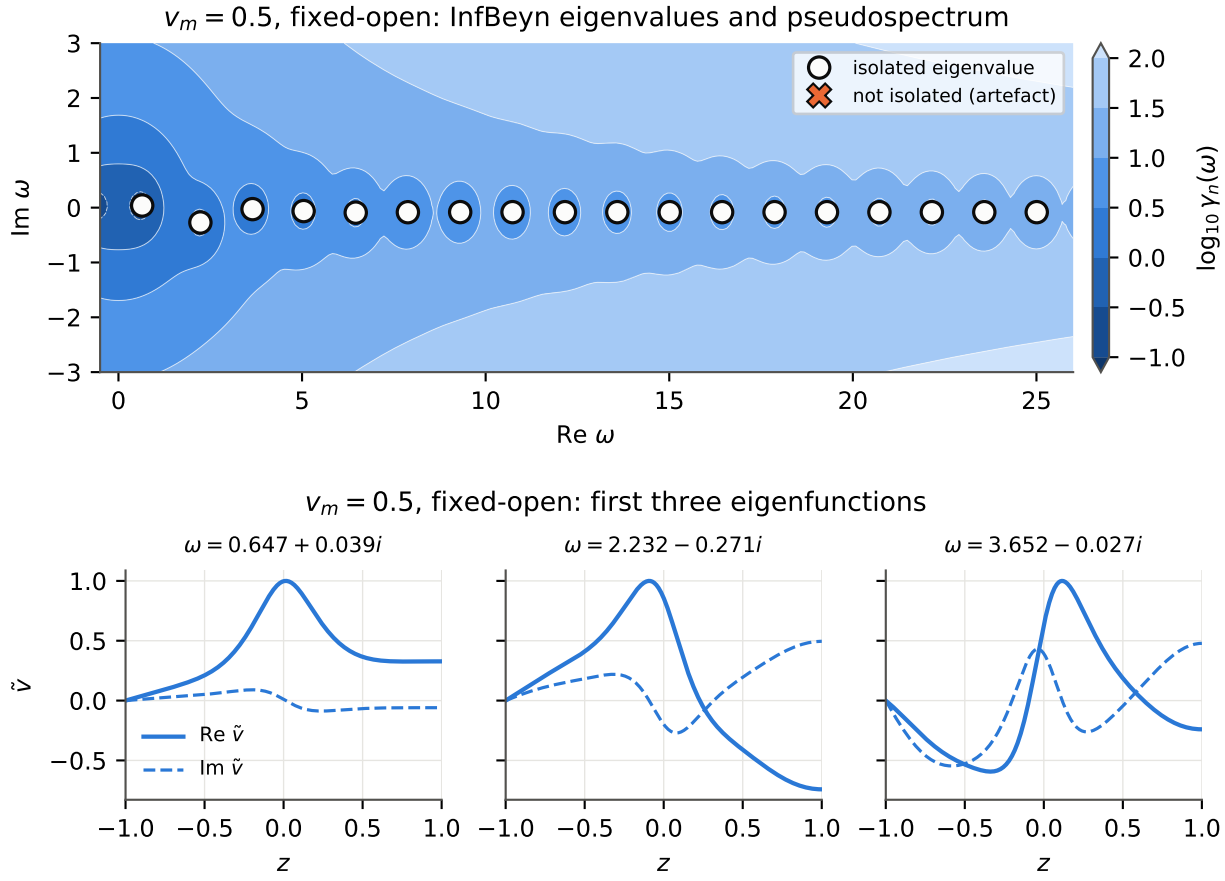


Figure 7: Varying profile, $v_m = 0.5$, fixed-open. Only the ground mode has $\mathrm{Im}\,\omega > 0$.

n	ω_n (InfBeyn)	$\gamma_{256}(\omega_n)$	halo	polished
1	$0.64661598 + 0.03884264i$	1.4×10^{-9}	2.5×10^{-1}	yes
2	$2.23164113 - 0.27090622i$	2.2×10^{-9}	9.3×10^{-1}	yes
3	$3.65213616 - 0.02712519i$	1.6×10^{-9}	1.6×10^{0}	yes
4	$5.04410886 - 0.06304891i$	1.4×10^{-9}	2.3×10^{0}	yes
5	$6.46204378 - 0.09108792i$	1.5×10^{-9}	3.0×10^{0}	yes
6	$7.88229735 - 0.08332761i$	1.4×10^{-9}	3.7×10^{0}	yes
7	$9.30667567 - 0.08135647i$	1.6×10^{-9}	4.4×10^{0}	yes
8	$10.73237805 - 0.08227976i$	1.6×10^{-9}	5.0×10^{0}	yes
9	$12.15874297 - 0.08246613i$	1.5×10^{-9}	5.7×10^{0}	yes
10	$13.58561766 - 0.08240018i$	1.5×10^{-9}	6.4×10^{0}	yes

Table 7: $v_m = 0.5$, fixed-open: first ten eigenvalues.

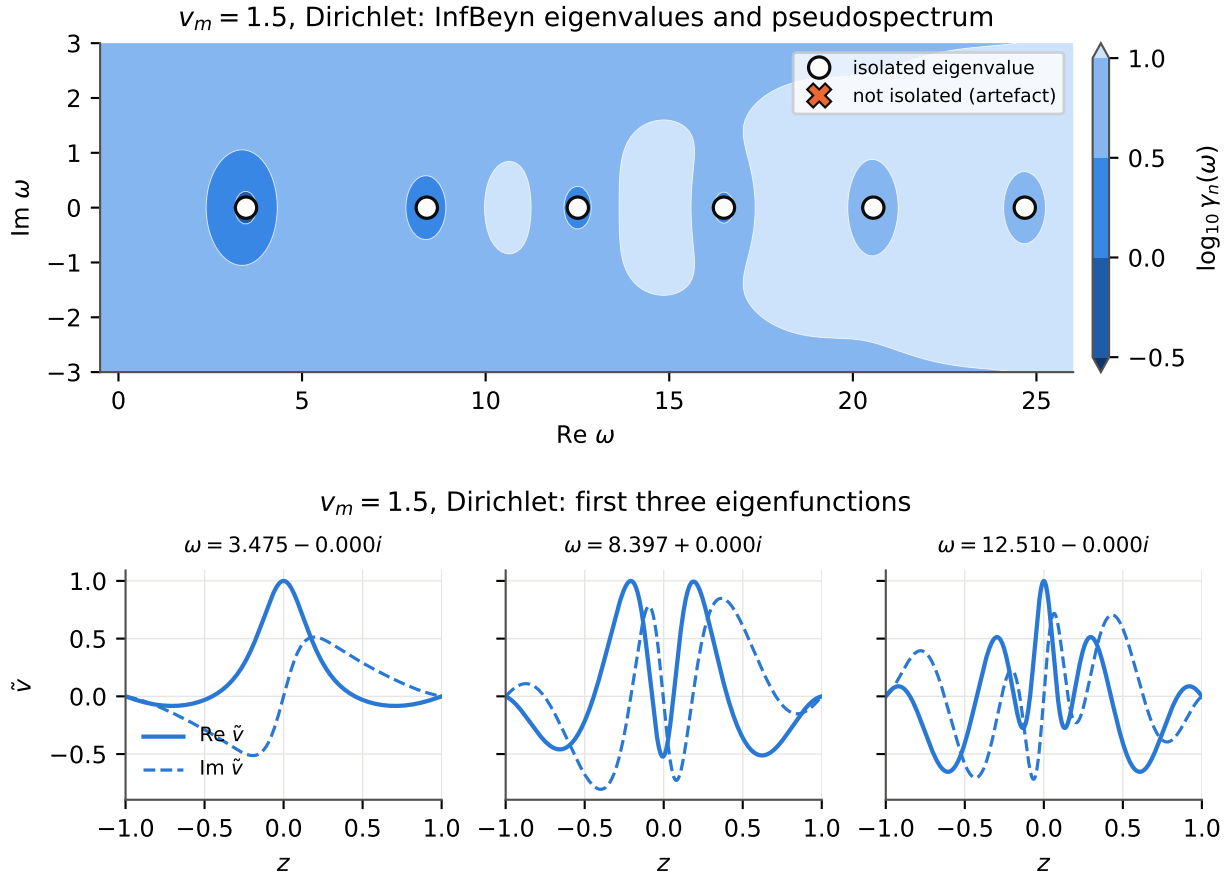


Figure 8: Varying profile, $v_m = 1.5$, Dirichlet.

n	ω_n (InfBeyn)	$\gamma_{256}(\omega_n)$	halo	polished
1	$3.47455636 - 0.00000000i$	6.3×10^{-9}	8.2×10^{-1}	yes
2	$8.39673137 + 0.00000000i$	5.2×10^{-9}	1.4×10^{0}	yes
3	$12.51019598 - 0.00000000i$	6.4×10^{-9}	2.1×10^{0}	yes
4	$16.48904036 + 0.00000000i$	6.1×10^{-9}	2.8×10^{0}	yes
5	$20.55778685 + 0.00000000i$	6.5×10^{-9}	3.6×10^{0}	yes
6	$24.68263122 + 0.00000000i$	6.8×10^{-9}	4.3×10^{0}	yes

Table 8: $v_m = 1.5$, Dirichlet: all six eigenvalues in the window.

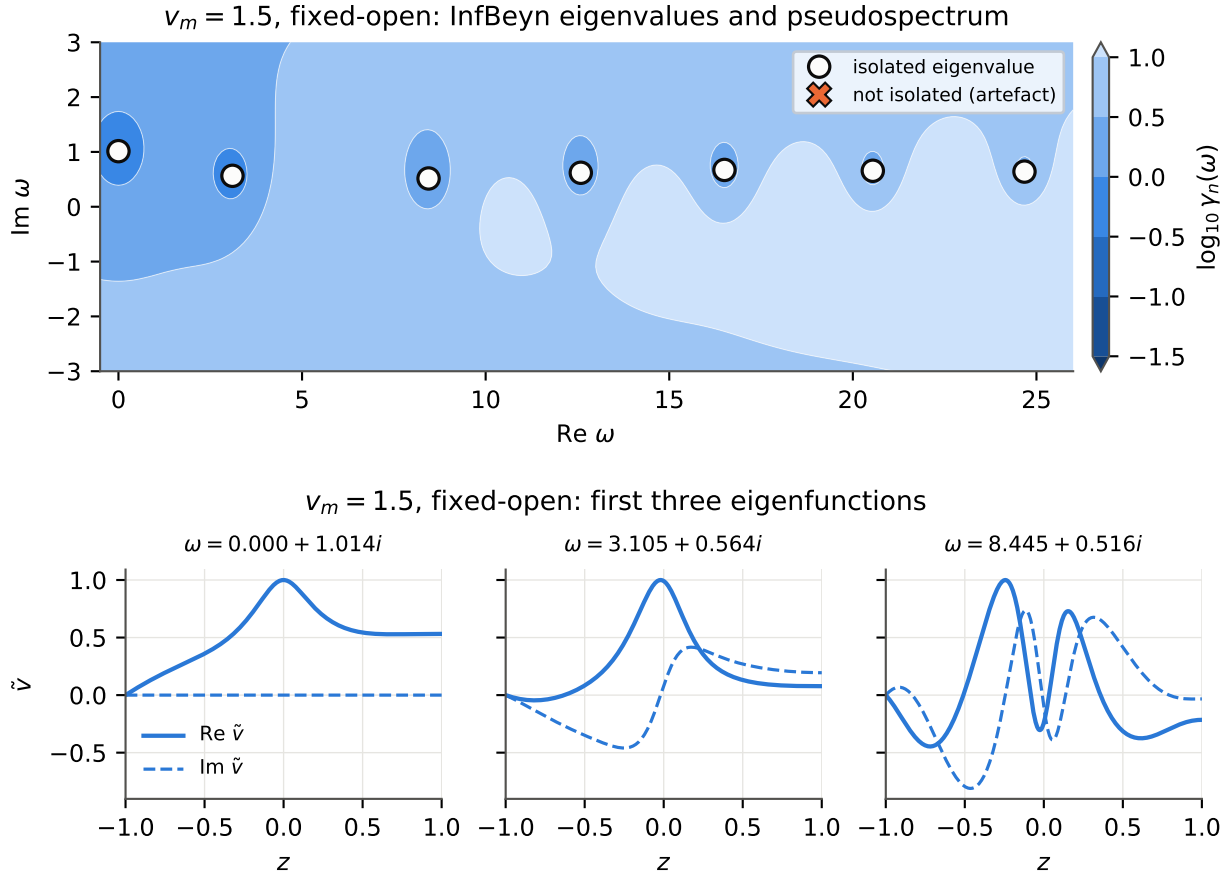


Figure 9: Varying profile, $v_m = 1.5$, fixed-open. Every mode is unstable.

n	ω_n (InfBeyn)	$\gamma_{256}(\omega_n)$	halo	polished
1	$0.00000001 + 1.01356972i$	3.1×10^{-8}	3.8×10^{-1}	no
2	$3.10504212 + 0.56428533i$	4.6×10^{-9}	5.2×10^{-1}	yes
3	$8.44537466 + 0.51585847i$	3.5×10^{-9}	1.2×10^{0}	yes
4	$12.59137405 + 0.61963297i$	4.6×10^{-9}	1.5×10^{0}	yes
5	$16.50345376 + 0.67161842i$	5.4×10^{-9}	1.9×10^{0}	yes
6	$20.54359191 + 0.65642344i$	5.3×10^{-9}	2.4×10^{0}	yes
7	$24.67418498 + 0.64081281i$	5.1×10^{-9}	2.9×10^{0}	yes

Table 9: $v_m = 1.5$, fixed-open: all seven eigenvalues in the window. See the text for the unpolished first mode.

4 Comparison with the discretize-then-eig methods

Figures 10–13 overlay, for each case, every eigenvalue of the finite-difference ($N = 101$, 201), collocation and Galerkin discretizations that falls in the scanned window on the InfBeyn eigenvalues, with a zoom on the imaginary parts of the first modes. Figures 14–15 show the distance from each InfBeyn eigenvalue to the nearest old eigenvalue, mode by mode, and for the constant-velocity cases the distance from InfBeyn to the exact value. Tables 11–14 list the same distances and the number of spurious eigenvalues; Table 10 collects the spurious counts.

case	BC	InfBeyn	FD N=101	FD N=201	collocation	Galerkin
constant $v_0 = 0.5$	Dirichlet	0	7	0	0	4
constant $v_0 = 0.5$	fixed-open	0	8	0	0	5
constant $v_0 = 1.5$	Dirichlet	0	7	7	0	2
constant $v_0 = 1.5$	fixed-open	0	8	7	0	1
$v_m = 0.5$	Dirichlet	0	2	0	0	6
$v_m = 0.5$	fixed-open	0	1	0	0	5
$v_m = 1.5$	Dirichlet	0	4	3	0	3
$v_m = 1.5$	fixed-open	0	4	3	0	3

Table 10: Number of eigenvalues in the window $\mathrm{Re}\,\omega \in [-0.5, 26]$, $\mathrm{Im}\,\omega \in [-3, 3]$ farther than 0.1 from every InfBeyn eigenvalue (spurious), per method.

Accuracy. For the constant-velocity problems the exact solution settles the comparison: InfBeyn and collocation both reach 10^{-13} to 10^{-12} on every mode in the window. Galerkin with 30 basis functions is as accurate for the first three modes and then loses one to two digits per mode (thesis Fig. 4.9, 4.13, 4.15), down to 10^{-1} by mode 8 for $v_0 = 1.5$. Finite differences are second order and their error grows like the square of the wavenumber: 10^{-3} on the first mode and 10^{-1} to 10^{0} by mode 7 at $N = 101$, four times smaller at $N = 201$, which is consistent with the dispersion analysis of thesis Sec. 4.4.1. For the varying profiles there is no exact solution, and InfBeyn is the most accurate of the four, so the distances in Tables 13, 14 are the errors of the old methods. Collocation at $N = 101$ agrees with InfBeyn to 10^{-12}–10^{-10} with Dirichlet boundaries for both profiles, but only to 10^{-3} with fixed-open boundaries (1.2×10^{-3} on the first subsonic mode, 3.5×10^{-3} on the first supersonic mode, 10^{-4} beyond). The loss is not a property of the collocation method but of the thesis-era script (`archive/run_experiments.py`): for the fixed-open case it rewrites one row of the Chebyshev differentiation matrices in place to impose $\tilde{v}'(1) = 0$, and then differentiates the equilibrium profile with the same, already modified, matrices, so v_0' and v_0'' are wrong at the grid point next to $z = 1$. Constant profiles (zero derivatives) and Dirichlet runs (no row rewriting)

are unaffected, which is why those agree to 10^{-12}. Repeating the computation with copies of the matrices reproduces the stored eigenvalues to 10^{-9} when the aliasing is kept and brings the collocation eigenvalues to within 5×10^{-8} ($v_m = 1.5$) and 10^{-9} ($v_m = 0.5$) of the InfBeyn values when it is removed. The thesis statement that collocation and Galerkin "agree well on the first three modes and start to deviate afterwards" for $v_m = 1.5$ (thesis Figs. 4.18, 4.19) is explained by Tables 14: the collocation eigenvalues stay within 10^{-10} (Dirichlet) or 10^{-3} (fixed-open) of the certified ones for all modes, while the Galerkin ones drift to 10^{-2} at mode 3 and to 10^{0} at mode 5. The Galerkin values, not the collocation ones, were the inaccurate ones.

Spurious eigenvalues. Every discretization produces eigenvalues that are not eigenvalues of the operator. Most of them have $|\operatorname{Im}\omega|$ of order 10 to 10^3 and fall outside the window, but Table 10 shows that several remain inside it: up to 8 for finite differences at $N = 101$ and up to 6 for Galerkin, in every one of the eight cases. In the thesis they were removed by comparing two resolutions (Sec. 4.4.2). InfBeyn produces none, because every eigenvalue it returns is a pole of the resolvent inside a contour, confirmed by the rank gap, the nested quadrature rule, Newton polishing and γ_n. Collocation at $N = 101$ shows no spurious eigenvalue in the window for any case, but its eigenvalues off the window (e.g. $\pm 2245i$ for constant $v_0 = 1.5$) show that this is a property of the window rather than of the method.

Completeness. The scan also answers a question the old approach cannot: that there are no further eigenvalues in the window. The pseudospectral plots are uniformly light away from the eigenvalues (no hidden region of small γ_n), and the accepted tiles cover the entire window.

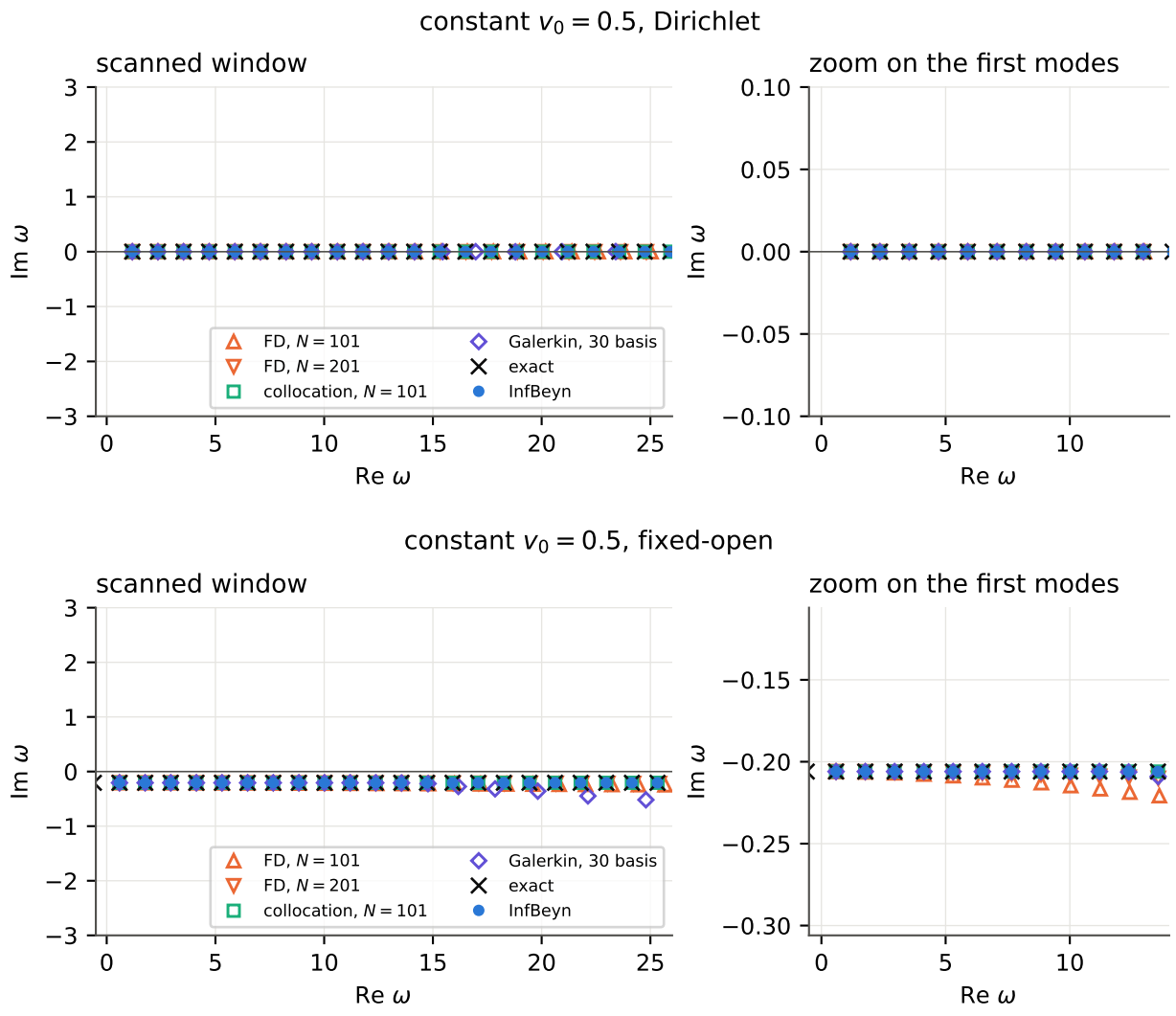


Figure 10: Constant $v_0 = 0.5$: all eigenvalues of the old discretizations in the window, with InfBeyn and exact values.

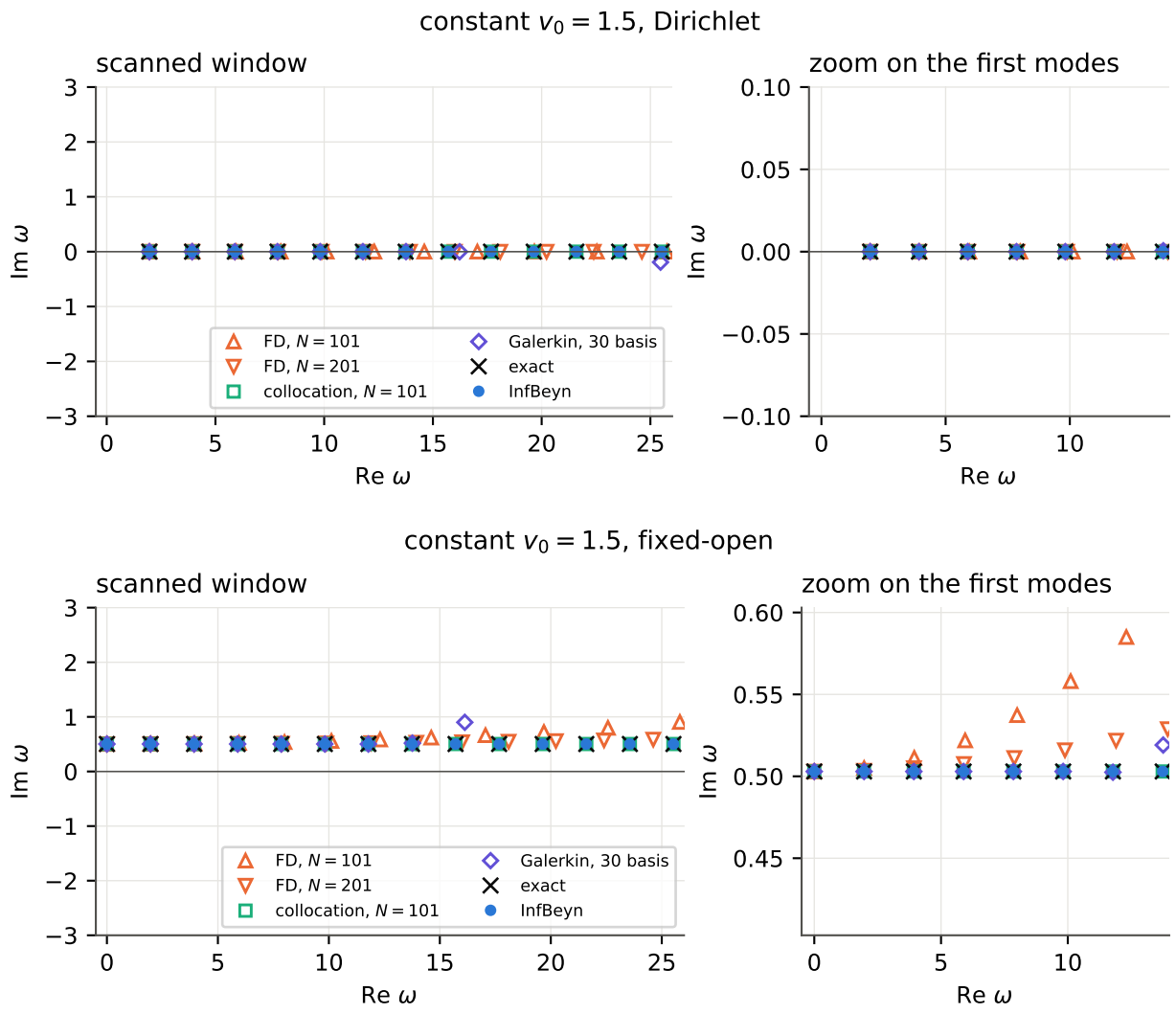


Figure 11: Constant $v_0 = 1.5$.

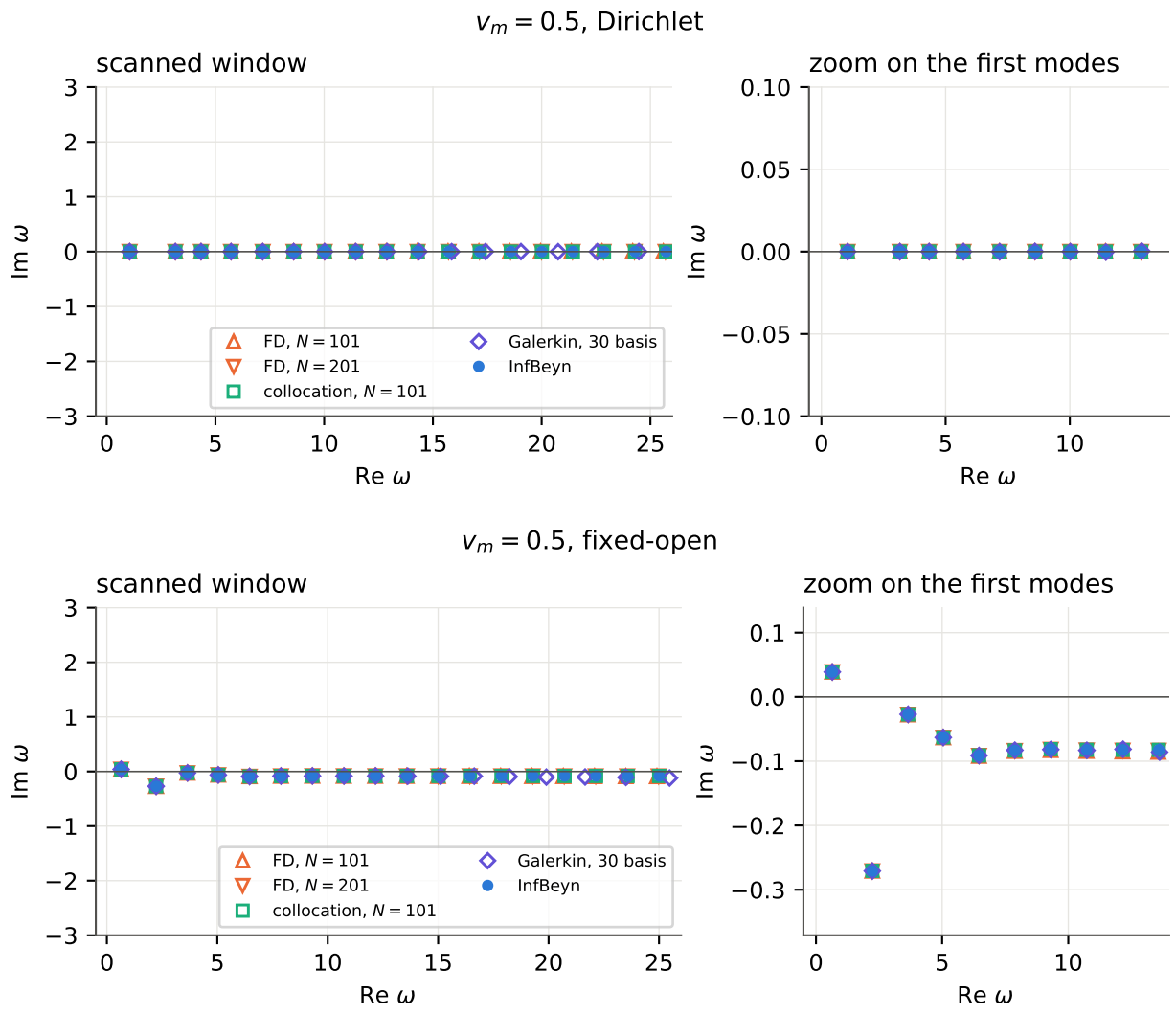


Figure 12: Varying profile, $v_m = 0.5$.

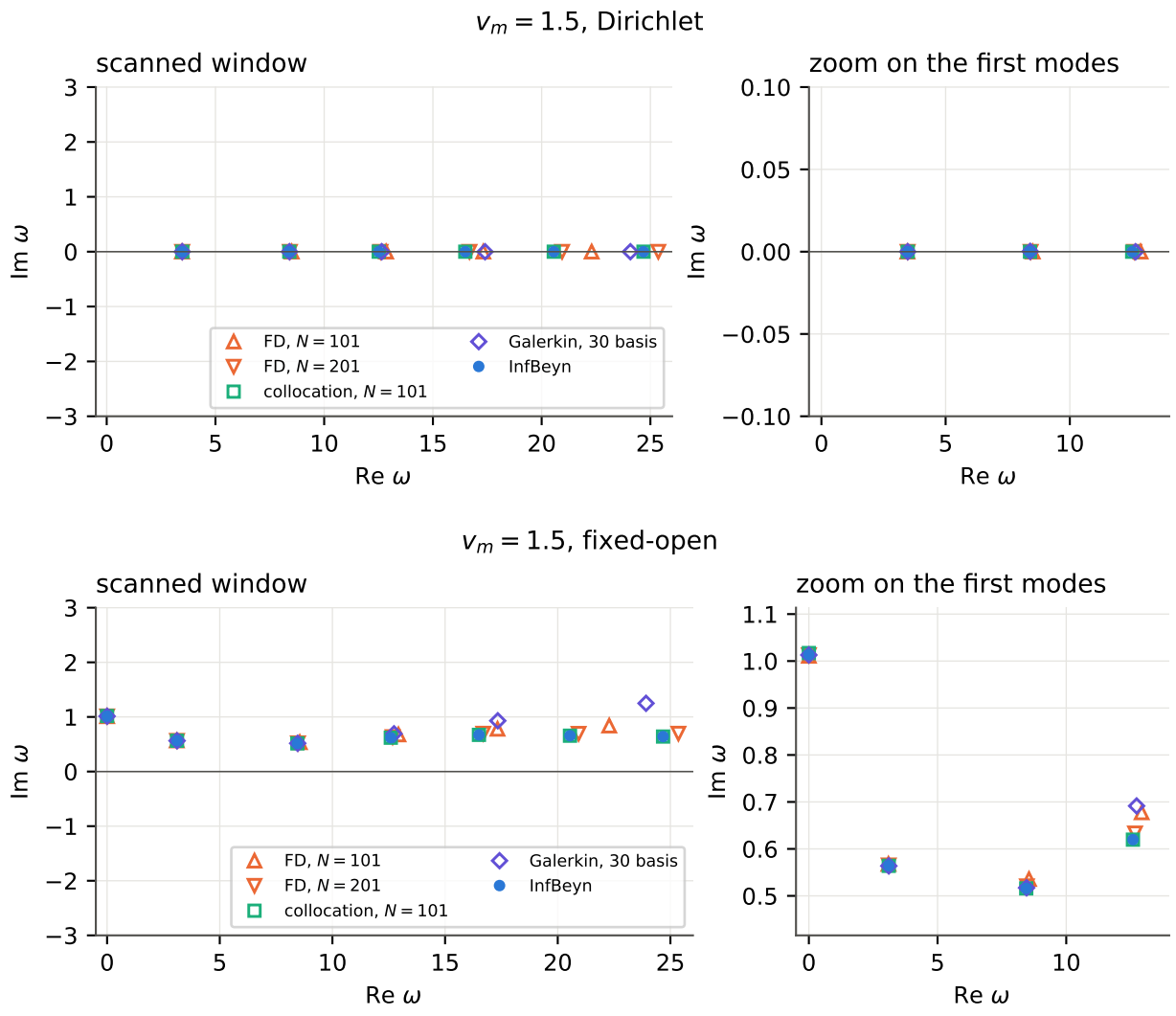


Figure 13: Varying profile, $v_m = 1.5$.

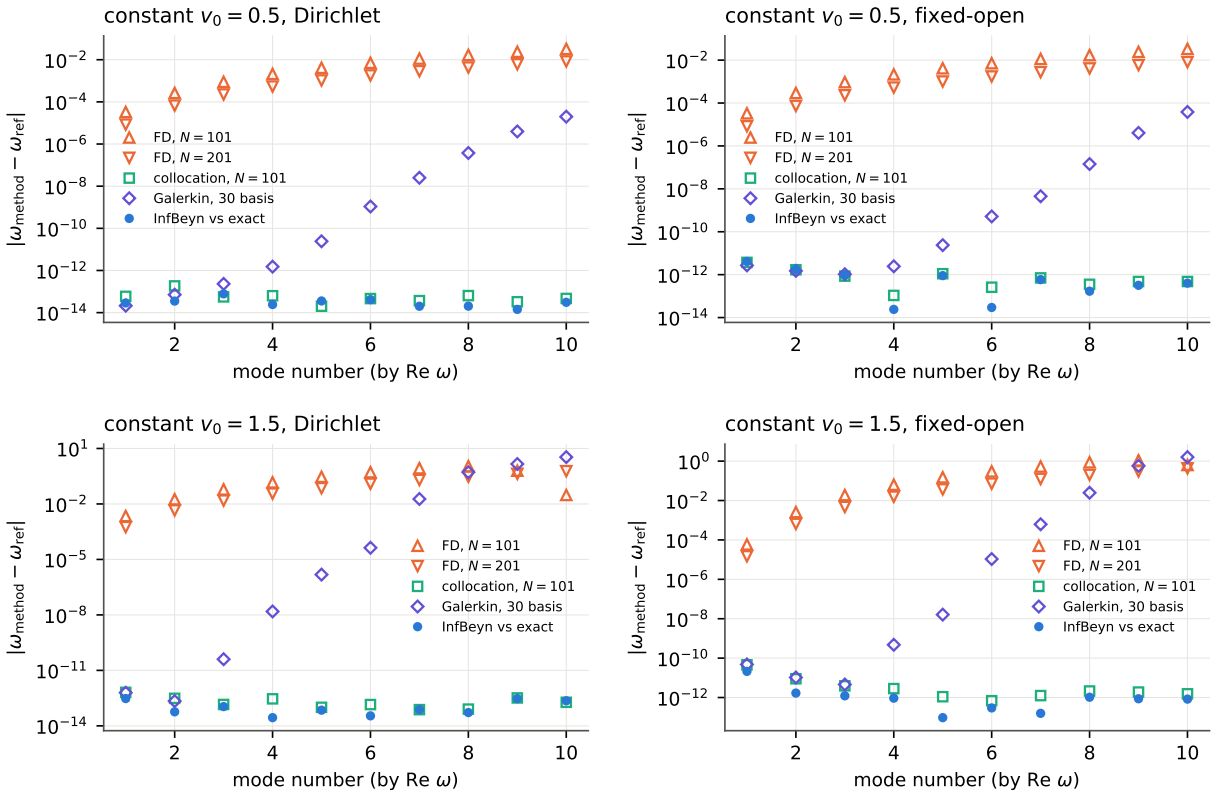


Figure 14: Constant velocity: distance from each InfBeyn eigenvalue to the nearest eigenvalue of each old method (open markers) and to the exact eigenvalue (filled circles), by mode number.

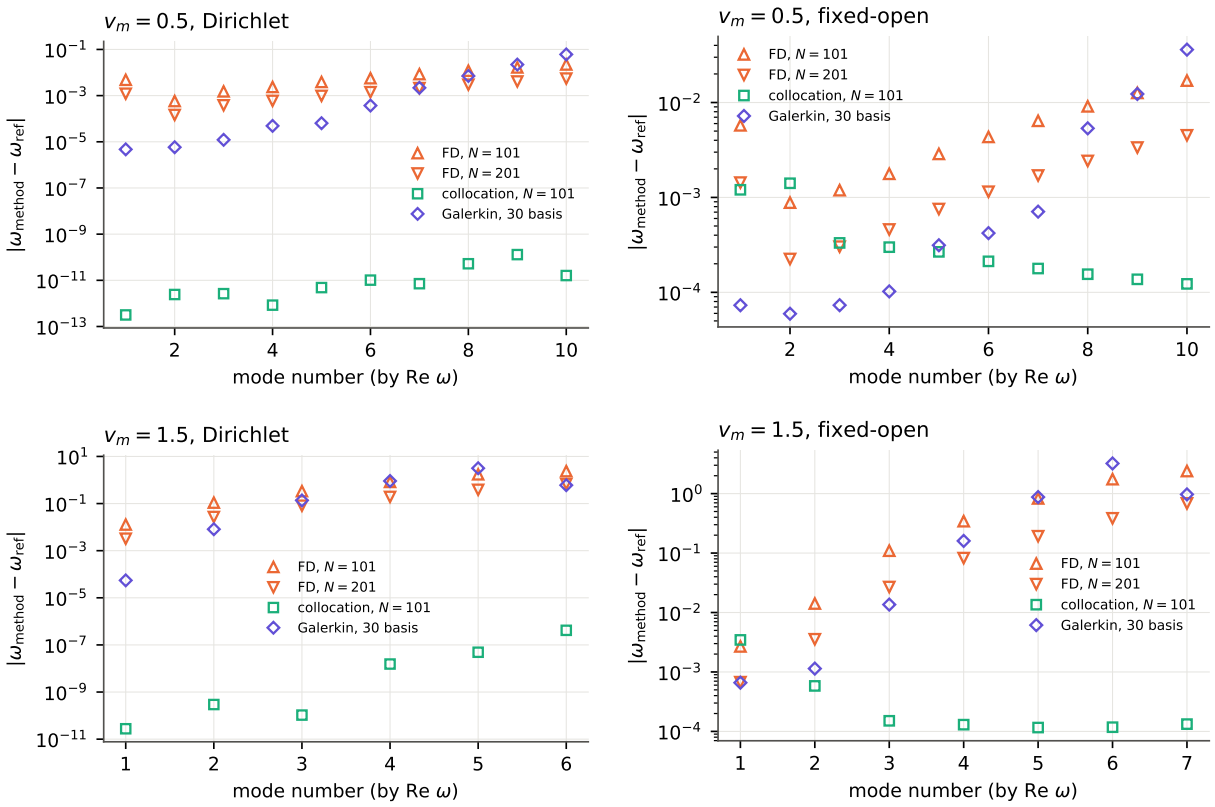


Figure 15: Varying profiles: distance from each InfBeyn eigenvalue to the nearest eigenvalue of each old method.

n	FD, $N = 101$	FD, $N = 201$	collocation, $N = 101$	Galerkin, 30 basis
1	3.3×10^{-5}	8.3×10^{-6}	5.9×10^{-14}	2.1×10^{-14}
2	2.7×10^{-4}	6.7×10^{-5}	1.9×10^{-13}	7.1×10^{-14}
3	9.0×10^{-4}	2.2×10^{-4}	5.6×10^{-14}	2.3×10^{-13}
4	2.1×10^{-3}	5.3×10^{-4}	6.3×10^{-14}	1.5×10^{-12}
5	4.1×10^{-3}	1.0×10^{-3}	2.0×10^{-14}	2.4×10^{-11}
6	7.2×10^{-3}	1.8×10^{-3}	4.6×10^{-14}	1.1×10^{-9}
7	1.1×10^{-2}	2.9×10^{-3}	3.7×10^{-14}	2.5×10^{-8}
8	1.7×10^{-2}	4.3×10^{-3}	6.4×10^{-14}	3.8×10^{-7}
in window	21	21	22	17
spurious	7	0	0	4

n	FD, $N = 101$	FD, $N = 201$	collocation, $N = 101$	Galerkin, 30 basis
1	3.3×10^{-5}	8.2×10^{-6}	3.7×10^{-12}	2.7×10^{-12}
2	3.0×10^{-4}	7.3×10^{-5}	1.7×10^{-12}	1.5×10^{-12}
3	9.4×10^{-4}	2.3×10^{-4}	8.6×10^{-13}	1.0×10^{-12}
4	2.2×10^{-3}	5.2×10^{-4}	1.1×10^{-13}	2.4×10^{-12}
5	4.2×10^{-3}	1.0×10^{-3}	1.1×10^{-12}	2.4×10^{-11}
6	7.2×10^{-3}	1.7×10^{-3}	2.6×10^{-13}	5.2×10^{-10}
7	1.1×10^{-2}	2.7×10^{-3}	7.1×10^{-13}	4.5×10^{-9}
8	1.7×10^{-2}	4.1×10^{-3}	3.5×10^{-13}	1.4×10^{-7}
in window	22	22	22	18
spurious	8	0	0	5

Table 11: Constant $v_0 = 0.5$, Dirichlet (left) and fixed-open (right): $|\omega_n^{\text{method}} - \omega_n^{\text{InfBeyn}}|$ for the first eight modes, number of old eigenvalues in the window and number of spurious ones.

n	FD, $N = 101$	FD, $N = 201$	collocation, $N = 101$	Galerkin, 30 basis
1	2.2×10^{-3}	5.6×10^{-4}	6.5×10^{-13}	6.2×10^{-13}
2	1.8×10^{-2}	4.5×10^{-3}	3.1×10^{-13}	2.2×10^{-13}
3	6.2×10^{-2}	1.5×10^{-2}	1.4×10^{-13}	4.0×10^{-11}
4	1.5×10^{-1}	3.6×10^{-2}	2.9×10^{-13}	1.5×10^{-8}
5	3.0×10^{-1}	7.1×10^{-2}	1.0×10^{-13}	1.5×10^{-6}
6	5.2×10^{-1}	1.2×10^{-1}	1.4×10^{-13}	4.3×10^{-5}
7	8.6×10^{-1}	2.0×10^{-1}	7.6×10^{-14}	1.9×10^{-2}
8	1.1×10^{0}	3.0×10^{-1}	8.0×10^{-14}	5.2×10^{-1}
in window	11	12	13	9
spurious	7	7	0	2

n	FD, $N = 101$	FD, $N = 201$	collocation, $N = 101$	Galerkin, 30 basis
1	5.9×10^{-5}	1.5×10^{-5}	4.6×10^{-11}	4.8×10^{-11}
2	2.6×10^{-3}	6.5×10^{-4}	8.9×10^{-12}	1.0×10^{-11}
3	1.9×10^{-2}	4.7×10^{-3}	3.9×10^{-12}	4.6×10^{-12}
4	6.4×10^{-2}	1.6×10^{-2}	2.8×10^{-12}	4.7×10^{-10}
5	1.5×10^{-1}	3.7×10^{-2}	1.1×10^{-12}	1.6×10^{-8}
6	3.0×10^{-1}	7.2×10^{-2}	6.9×10^{-13}	1.1×10^{-5}
7	5.3×10^{-1}	1.2×10^{-1}	1.3×10^{-12}	6.2×10^{-4}
8	8.8×10^{-1}	2.0×10^{-1}	2.1×10^{-12}	2.5×10^{-2}
in window	12	13	14	9
spurious	8	7	0	1

Table 12: Constant $v_0 = 1.5$, Dirichlet (left) and fixed-open (right).

n	FD, $N = 101$	FD, $N = 201$	collocation, $N = 101$	Galerkin, 30 basis
1	4.8×10^{-3}	1.2×10^{-3}	3.2×10^{-13}	4.7×10^{-6}
2	5.8×10^{-4}	1.5×10^{-4}	2.4×10^{-12}	5.8×10^{-6}
3	1.5×10^{-3}	3.8×10^{-4}	2.6×10^{-12}	1.2×10^{-5}
4	2.4×10^{-3}	5.9×10^{-4}	8.4×10^{-13}	4.9×10^{-5}
5	3.9×10^{-3}	9.7×10^{-4}	4.8×10^{-12}	6.4×10^{-5}
6	5.7×10^{-3}	1.4×10^{-3}	1.0×10^{-11}	3.7×10^{-4}
7	8.5×10^{-3}	2.1×10^{-3}	7.2×10^{-12}	2.2×10^{-3}
8	1.2×10^{-2}	3.0×10^{-3}	5.2×10^{-11}	7.1×10^{-3}
in window	18	18	18	16
spurious	2	0	0	6

n	FD, $N = 101$	FD, $N = 201$	collocation, $N = 101$	Galerkin, 30 basis
1	5.8×10^{-3}	1.4×10^{-3}	1.2×10^{-3}	7.3×10^{-5}
2	8.8×10^{-4}	2.2×10^{-4}	1.4×10^{-3}	6.0×10^{-5}
3	1.2×10^{-3}	3.0×10^{-4}	3.3×10^{-4}	7.3×10^{-5}
4	1.8×10^{-3}	4.6×10^{-4}	3.0×10^{-4}	1.0×10^{-4}
5	2.9×10^{-3}	7.5×10^{-4}	2.7×10^{-4}	3.1×10^{-4}
6	4.3×10^{-3}	1.1×10^{-3}	2.1×10^{-4}	4.2×10^{-4}
7	6.4×10^{-3}	1.7×10^{-3}	1.8×10^{-4}	7.1×10^{-4}
8	9.1×10^{-3}	2.4×10^{-3}	1.6×10^{-4}	5.3×10^{-3}
in window	18	18	18	17
spurious	1	0	0	5

Table 13: Varying $v_m = 0.5$, Dirichlet (left) and fixed-open (right).

n	FD, $N = 101$	FD, $N = 201$	collocation, $N = 101$	Galerkin, 30 basis
1	1.3×10^{-2}	3.2×10^{-3}	2.8×10^{-11}	5.5×10^{-5}
2	1.1×10^{-1}	2.7×10^{-2}	3.0×10^{-10}	8.1×10^{-3}
3	3.4×10^{-1}	8.1×10^{-2}	1.1×10^{-10}	1.4×10^{-1}
4	8.3×10^{-1}	1.9×10^{-1}	1.5×10^{-8}	9.1×10^{-1}
5	1.7×10^{0}	3.8×10^{-1}	4.9×10^{-8}	3.2×10^{0}
6	2.4×10^{0}	6.8×10^{-1}	4.2×10^{-7}	6.0×10^{-1}
in window	5	6	6	5
spurious	4	3	0	3

n	FD, $N = 101$	FD, $N = 201$	collocation, $N = 101$	Galerkin, 30 basis
1	2.7×10^{-3}	6.7×10^{-4}	3.5×10^{-3}	6.6×10^{-4}
2	1.4×10^{-2}	3.5×10^{-3}	5.8×10^{-4}	1.1×10^{-3}
3	1.1×10^{-1}	2.7×10^{-2}	1.5×10^{-4}	1.4×10^{-2}
4	3.4×10^{-1}	8.2×10^{-2}	1.3×10^{-4}	1.6×10^{-1}
5	8.3×10^{-1}	1.9×10^{-1}	1.2×10^{-4}	8.7×10^{-1}
6	1.8×10^{0}	3.8×10^{-1}	1.2×10^{-4}	3.2×10^{0}
7	2.4×10^{0}	6.8×10^{-1}	1.3×10^{-4}	9.7×10^{-1}
in window	6	7	7	6
spurious	4	3	0	3

Table 14: Varying $v_m = 1.5$, Dirichlet (left) and fixed-open (right).

5 Why only the ground mode of the subsonic fixed-open flow is unstable

The subsonic ($v_m = 0.5$) fixed-open spectrum has one growing mode, $\omega_1 = 0.6466 + 0.0388i$, while every other mode is damped. Three computations (`investigate_subsonic_open.py`) explain this.

Energy budget. Multiplying Eq. (1) by $\bar{\tilde{v}}$, integrating over $[-1, 1]$ and using $\tilde{v}(-1) = \tilde{v}'(1) = 0$ gives $a\omega^2 + b\omega + c = 0$ with $a = \int |\tilde{v}|^2$ real. The imaginary part of this relation is

$$\operatorname{Im}\omega = -\frac{\operatorname{Re}\omega\, v_0(1)|\tilde{v}(1)|^2 + \operatorname{Re}\omega \int v_0'|\tilde{v}|^2 - \operatorname{Im}\int (v_0 + 1/v_0)\, v_0'\, \bar{\tilde{v}}\tilde{v}'}{2\operatorname{Re}\omega\, a + \operatorname{Re} b}, \qquad \operatorname{Re} b = -2\operatorname{Im}\int v_0 \bar{\tilde{v}}\tilde{v}', \quad (6)$$

where the three terms of the numerator are, in order, the radiation of energy through the open end (always damping, as it is the only term left for constant v_0 and gives $\operatorname{Im}\omega = -0.206$ there), the work of the velocity gradient on the mode, and the convective coupling through v_0'. The denominator is positive for all modes. Evaluated with the certified eigenfunctions, Eq. (6) reproduces every $\operatorname{Im}\omega$ to 10^{-12}. Figure 16 shows the three contributions. For the ground mode the radiation term is tiny (-0.0003 before dividing by the denominator) because $\operatorname{Re}\omega$ is small and the mode is nearly flat at the exit, while both gradient terms are positive ($+0.0005$ each); the mode gains energy from the flow gradient faster than it radiates it. For mode 2 all three terms are negative and large, and for every higher mode the radiation and coupling terms are negative and grow with $\operatorname{Re}\omega$, while the gradient term is small or negative. Figure 17 shows why: $|\tilde{v}_1|^2$ is concentrated on the accelerating flank $z < 0$ where $v_0' > 0$, and is small at the exit, whereas $|\tilde{v}_2|^2$ straddles both flanks and is large at the open end.

Continuation in the profile amplitude. Following the six lowest modes with Chebyshev collocation ($N = 200$) as the mirror ratio R is reduced from 1.5 to 0 with the throat velocity held at $v_m = 0.5$ (Fig. 18, left), all six converge to the constant-velocity values $\omega_n = (v_0^2 - 1)[(2n+1)\pi/4 - \frac{i}{4}\ln\frac{1}{3}]$ with the common damping -0.206. As the bump grows the exit velocity $v_0(\pm 1)$ falls (to 0.19 at $R = 1.5$), the radiation term, proportional to $v_0(1)$, weakens, and every mode becomes less damped; the limit $\operatorname{Im}\omega \to -0.0825$ of the high modes is the constant-velocity damping rate evaluated at the exit velocity, -0.088. The ground mode is the one that in addition draws energy from v_0', and it crosses $\operatorname{Im}\omega = 0$ at $R \approx 0.74$. The instability is therefore a continuous deformation of the lowest damped mode of the uniform flow, not a new branch.

Continuation in the throat Mach number. At $R = 1.5$ the ground mode is unstable for every v_m from 0.1 to 0.95, with a growth rate increasing monotonically with v_m (Fig. 18, right); modes 2 and 3 stay damped. The result is therefore robust, not a coincidence of $v_m = 0.5$.

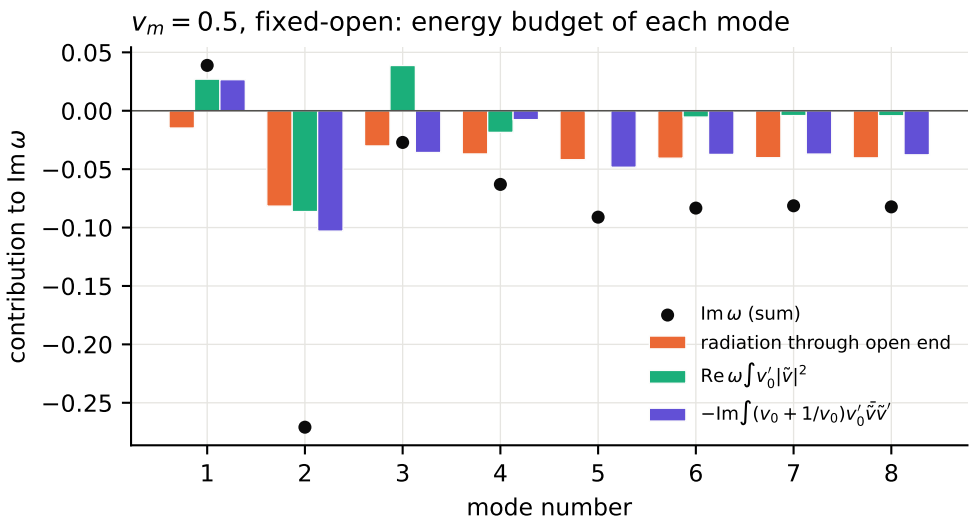


Figure 16: Energy budget, Eq. (6): the three contributions to $\mathrm{Im}\,\omega$ for the first eight modes of the $v_m = 0.5$ fixed-open flow. Their sum (dots) equals the computed growth rate.

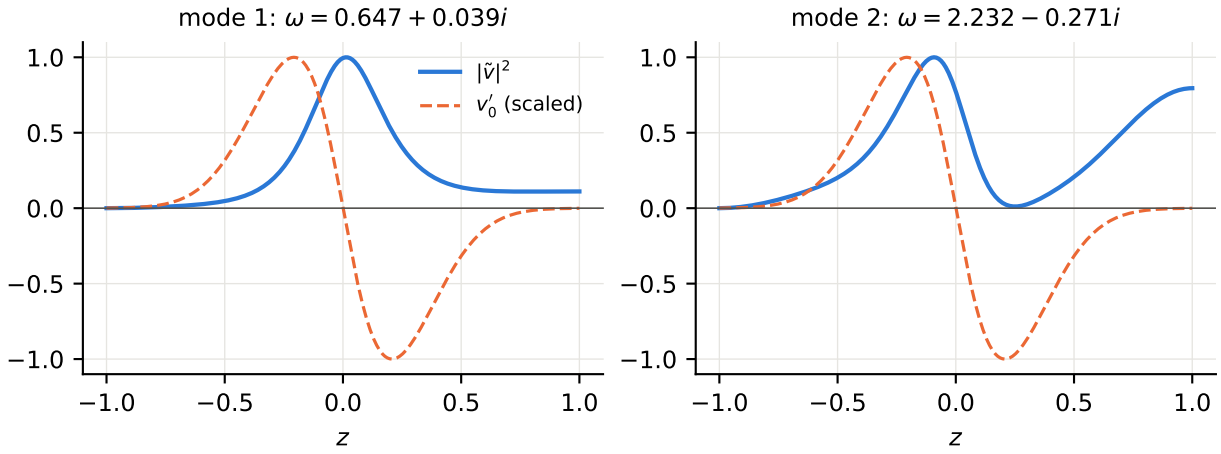


Figure 17: $|\tilde{v}|^2$ of the first two modes against the velocity gradient v_0'. The unstable ground mode lives on the accelerating flank and is small at the open end.

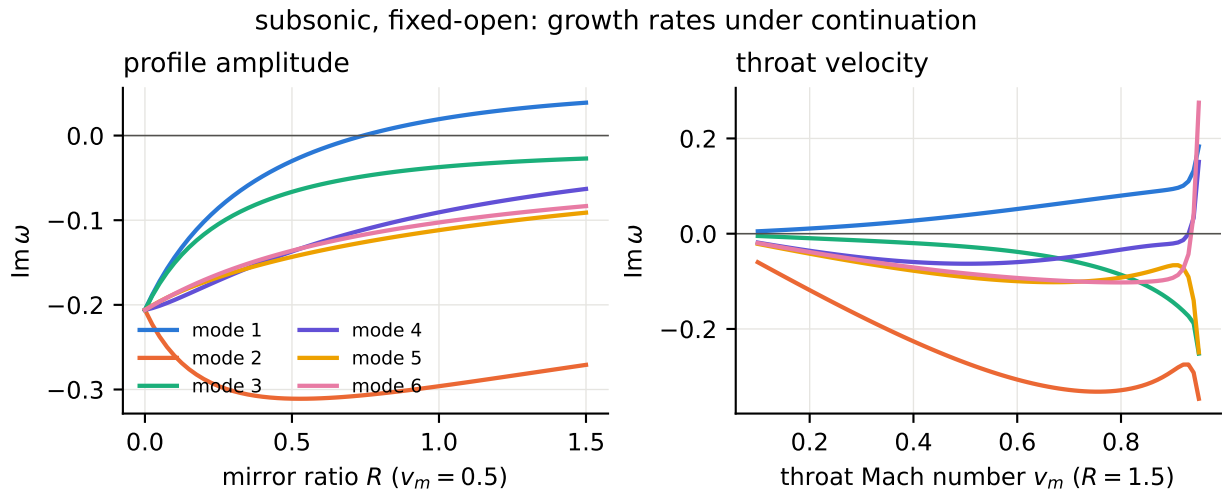


Figure 18: Growth rates of the six lowest modes as the profile amplitude R (left, $v_m = 0.5$) and the throat Mach number v_m (right, $R = 1.5$) are varied.

6 Conclusions

- InfBeyn reproduces all stability verdicts of the thesis for the non-transonic flows, now with a certificate for each eigenvalue and a guarantee that no eigenvalue in the window was missed. No convergence test or spurious-mode filtering is needed.
- For constant velocity the eigenvalues agree with the exact ones to 10^{-13} (to 10^{-11} for the purely imaginary fixed-open supersonic mode). The thesis-era collocation is equally accurate there; finite differences and Galerkin are not.
- For the varying profiles the collocation results of the thesis are accurate to 10^{-10} with Dirichlet and, because of an aliasing slip in the old script, only to 10^{-3} with fixed-open boundaries (to 10^{-8} once the slip is removed); the disagreement between collocation and Galerkin above the third mode reported in the thesis is due to the Galerkin values.
- Thesis Eq. (3.12) should read with phase $(2n + 1)\pi/4$ for $v_0 < 1$; the figures of the thesis already use the correct values.
- Certified growth rates: 0.50295 (constant $v_0 = 1.5$, fixed-open, all modes), 0.03884 (varying $v_m = 0.5$, fixed-open, ground mode only) and 1.01357 (varying $v_m = 1.5$, fixed-open, $\mathrm{Re}\,\omega = 0$ mode).

Reproduction. From the repository root: `uv run python nontransonic_flow/nozzle_infbeyn.py` (about 2.5 minutes on 12 cores, writes `nontransonic_flow/data/infbeyn_*.npz`), then `uv run python nontransonic_flow/make_figures.py` and `tectonic nontransonic_flow/report.tex`.

References

[1] H. Feng, *Spectral stability of plasma flows in the magnetic nozzle*, M.Sc. thesis, University of Saskatchewan, 2024.

[2] M. J. Colbrook and A. Townsend, *Avoiding discretization issues for nonlinear eigenvalue problems*, arXiv:2305.01691, 2023.